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CS 499: Senior Project
Teacher: Rob Tuft
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Project Proposal

Project Name

Home Gig

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Project Purpose

Home-run businesses like small farms often rely on disconnected and low-technology workflows to manage their daily operations. Customer orders may come from multiple sources such as websites, phone calls, text messages, social media, or in-person conversations. Home run businesses frequently track these orders manually using paper notes or spreadsheets, making it difficult to organize inventory, monitor deliveries, reconcile payments, and maintain accurate records. This can lead to missed orders, inconsistent payment tracking, duplicate data entry, and inefficient bookkeeping processes.

The proposed system, which I will call Home Gig, will solve this problem by creating a centralized order operations platform designed specifically for the needs of a small, run-from-home, business. To keep the project realistic for a solo developer while still producing a polished and complete product, the first release of Home Gig will focus on the owner-side operations dashboard rather than a public customer storefront. The core product will allow small business owners to manage products, customers, orders, inventory, payments, deliveries, and spreadsheet exports from one centralized system. A public storefront and customer online ordering capabilities may be added as stretch goals if time allows, but they are not required for the minimum completed product. The primary focus of the project will be an owner dashboard that unifies phone, text message, social media, and in-person orders into one manageable workflow. Business owners will also have the option to install the responsive web application as a Progressive Web App (PWA) on their mobile devices for easier access to order management, operational workflows, and business records.

The owner dashboard will focus on the operational needs of the business by allowing owners to manage products, track incoming orders, manually enter verbal or phone orders, record payments, manage deliveries, and export accounting-ready spreadsheet reports. The project emphasizes improving organization and simplifying real-world home-run business workflows rather than competing with large commercial e-commerce platforms. By centralizing order management and reducing reliance on disconnected manual processes, the system will help home-based businesses operate more efficiently and maintain cleaner business records.

Background/Prior Knowledge

I consider myself a Newbie when it comes to the topic of the topic of home-run businesses, small farms and their operational workflows such as order management and bookkeeping. I understand that small home-run businesses need ways to manage sales, payments, inventory, and business records, but I currently have very limited real-world experience in this area. One of the goals of this project is to better understand the operational challenges that small home-run businesses face and how software solutions can help simplify those workflows.

I also consider myself a beginner with some of the technologies I will be using because I previously built a web application using Angular, Node.js, Express, and MongoDB in my WDD 430 Web Full-Stack Development course. However, I will be a Newbie when it comes to creating Progressive Web Application (PWA), because I currently have no prior experience developing installable web applications or implementing service workers and offline functionality. I also have no prior experience implementing downloadable spreadsheet exports inside a web application. Creating the functionality that will allow orders and payment records to be exported into spreadsheet-compatible formats such as CSV or Excel files will therefore be a completely new learning experience for me.

Although other applications exist that provide order management and business management services, my idea was not inspired by or built off of another application. My idea simply came from trying to solve the bookkeeping and organization struggles that my sister-in-law faced having to track orders and money while selling items from home that she naturally produced by doing the things she loved to do, namely, raising chickens, keeping bees, and gardening. My project differentiates itself from existing farm commerce and business platforms that primarily focus on digital storefronts and online customer ordering by emphasizing low-technology business operations management rather than primarily focusing on eCommerce functionality. My app will focus on simplifying the operational workflows of small home run businesses that manage orders from multiple low-technology sources such as phone calls, verbal requests, spreadsheets, and in-person sales.

The WDD 430 Web Full-Stack Development course is what originally sparked my interest in using the Angular, Node.js, Express, and MongoDB technology stack to create web applications. I plan to use the project I built in that course as a foundation to build upon while learning how to implement the additional features and technologies required for this project.

Description

Home Gig is a small-business operations platform designed for home-run businesses that manage orders through disconnected workflows such as phone calls, text messages, social media, spreadsheets, and in-person sales. The system centralizes customer orders, payments, inventory, deliveries, and business records into one simplified operational dashboard.

The initial demonstration use case focuses on a small local farm operation. The app will help my Sister-in-law's local home-run farm organize and centralize her daily business workflows. Many home businesses currently manage customer orders through disconnected systems such as handwritten notes, spreadsheets, text messages, phone calls, social media messages, and separate online ordering platforms. These disconnected workflows can lead to missed orders, duplicate data entry, inconsistent inventory tracking, billing inconsistencies, and inefficient bookkeeping processes.

The project will provide a centralized system with an owner-focused operational dashboard. The owner dashboard will allow the business owners to manage products, inventory, incoming orders, deliveries, and customer records within one unified workflow that can also be installed as a Progressive Web App (PWA) on mobile devices. If time allows, a public web storefront may be added as a stretch goal where customers can browse products, place online orders, and review order history through a responsive web application. However, the storefront is considered an optional enhancement rather than a required component of the minimum viable product.

A core feature of the system is support for multi-source order management. Business owners will be able to manually enter verbal, phone, text-message, and in-person orders directly into the system. This allows all customer orders to be tracked in one centralized location regardless of how the order was received. The system will also support payment tracking, delivery status management, and spreadsheet-ready export functionality for accounting and bookkeeping purposes.

Solutions similar to portions of this project already exist, including platforms such as Local Line, GrownBy, Homegrown, and Locally Grown. These systems primarily focus on eCommerce storefronts, CSA subscriptions, online customer ordering, and large integrated farm-commerce ecosystems. The proposed project differs by focusing specifically on simplified operational workflows for small home-run businesses that still rely heavily on manual and low-technology business processes. The system emphasizes owner-friendly management tools, verbal order entry, spreadsheet-first workflows, and simplified organization rather than attempting to compete with large commercial eCommerce platforms.

The project will be considered complete when a functional demonstration system is capable of performing the core operational workflows required by a small home-run business. A valid demo product will include:

- An owner dashboard for product and inventory management
- Manual entry of verbal, phone, text-message, social media, and in-person orders
- Centralized order tracking
- Payment and delivery status tracking
- Spreadsheet export functionality for accounting records
- A responsive mobile-friendly interface and installable PWA support

The project will be successful if it demonstrates a complete workflow where customer orders from multiple sources can be managed through one centralized operational system that simplifies organization and reduces reliance on disconnected manual processes.

Significance

This project is significant because it addresses a real operational problem faced by many small businesses run at home, including home-based agricultural businesses. Small farms and businesses often rely on disconnected manual workflows such as handwritten notes, spreadsheets, phone calls, text messages, and social media conversations to manage customer orders and daily operations. These fragmented systems can create inefficiencies, duplicate data entry, missed orders, inventory inconsistencies, and bookkeeping challenges. The proposed system provides real-world value by centralizing these workflows into a single operational platform that simplifies organization and improves business management.

The project has strong real-world applicability because it focuses on the practical operational needs of home-based businesses rather than only creating another online storefront. The system is designed to support workflows that many existing platforms do not emphasize, including manual verbal order entry, spreadsheet-ready accounting exports, and simplified low-technology operational management. This makes the project useful for small businesses that may not have advanced technical experience or the resources to adopt large enterprise eCommerce systems.

The project will apply multiple software development principles learned throughout the degree program, including:

- Full-stack web application development
- Database design and management
- REST API development
- User authentication and authorization
- Responsive user interface design
- Progressive Web Application (PWA) implementation
- Software architecture and modular design
- Version control and project management
- Testing and debugging practices

The project is also realistic and achievable within a semester because the scope is intentionally focused on a minimum viable operational platform rather than a large enterprise commerce ecosystem. Existing libraries, frameworks, and technologies will be used where appropriate to support efficient development and industry-standard practices.

This project would also be valuable to include on a professional resume because it demonstrates the ability to design and build a real-world full-stack business application with practical operational functionality. The project showcases experience with solving business workflow problems, designing user-focused operational systems, integrating multiple operational features into a unified platform, and developing a responsive application suitable for real-world use.

Here are some examples of resume statements I may use:

- Developed a full-stack home based business operations management platform using modern web technologies
- Designed a centralized order management system supporting verbal, phone, social media, and in-person sales workflows
- Built responsive Progressive Web Application (PWA) functionality for mobile-friendly business operations
- Implemented inventory management, payment tracking, delivery coordination, and spreadsheet export features
- Created RESTful APIs and database architecture to support multi-source order processing and operational reporting

The completed project would be demonstratable in professional interviews and portfolio presentations because it represents a realistic business application with clear real-world use cases, operational workflows, and user-focused problem solving.

New Computer Science Concepts

One of the major goals of this project is to further develop my ability to independently learn new software engineering concepts and technologies beyond those directly taught in previous coursework. While I already have beginner-level experience building a web application using Angular, Node.js, Express, and MongoDB from my WDD 430 Web Full-Stack Development course, this project will require me to expand my knowledge significantly and apply these technologies in more advanced and practical ways.

The existing project from WDD 430 will serve as a foundation that I can build upon while learning additional concepts required to create the Home Gig platform. Home Gig is intended to be a centralized operations and order management system for small home-run businesses that manage customer orders through disconnected workflows such as phone calls, text messages, social media messages, spreadsheets, and in-person sales.

One major new concept I will need to learn is Progressive Web Application (PWA) development. I currently have no prior experience creating PWAs. To support mobile-friendly business operations, I will need to learn how to configure Angular applications as installable PWAs, including concepts such as service workers, application manifests, offline caching strategies, install prompts, and responsive mobile behavior. This will require research into how modern web applications provide app-like functionality across mobile and desktop devices without requiring traditional app store deployment.

Another major area of learning will involve spreadsheet export and file download functionality. I currently have no experience implementing features that generate downloadable files within a web application. To complete this functionality, I will need to learn how to generate spreadsheet-compatible reports such as CSV or Excel files from backend order and payment data, as well as how to securely provide those files for download through the frontend application. This includes learning concepts related to server-side file generation, data formatting, file streaming, browser downloads, and report generation workflows. For spreadsheet export and downloadable reporting functionality, I expect to use libraries I have not used before such as ExcelJS or SheetJS to generate CSV and Excel-compatible files. I will also learn how to do file generation and spreadsheet export functionality with the use of ExcelJS or SheetJS. Additionally, I will learn a new programming language to me called Go to implement a small backend service for spreadsheet export and report generation.

The project will also require me to strengthen my understanding of full-stack software architecture and operational workflow design. Unlike previous coursework projects that focused on isolated features, this project involves coordinating multiple interconnected business operations including:

- Product management
- Inventory tracking
- Centralized order management
- Manual/verbal order entry
- Payment tracking
- Delivery management
- Customer management
- Reporting and export workflows

This will require learning more advanced approaches to database design, API organization, modular application architecture, state management, and operational workflow management.

Finally, the project will involve learning how to effectively use modern AI-assisted development tools and agentic workflow technologies as part of the software engineering process. I will be using the AI tools GitHub Copilot, ChatGPT, Codex, SuperGrok, and agent skills plugins to assist with research, troubleshooting, debugging, architectural planning, documentation, brainstorming, and code suggestions throughout development. These tools will be used similarly to technical documentation and developer support resources while ensuring that I maintain understanding of the implemented code, software engineering concepts, and overall system architecture. Using these tools for this project will allow me to learn how to responsibly integrate AI-assisted development and agentic workflow technologies into a real-world software project and apply them as modern software engineering skills.

Interestingness

This project is exciting to me personally because it involves several things I am very interested in. Those things are solving real-world problems through software development, helping someone I care about, and entrepreneurship. One of my long-term goals is to eventually develop, market, and manage my own applications, so this project provides an opportunity to gain practical experience building a realistic business-oriented software platform from the ground up. I am especially interested in creating applications that improve people's lives in personal and meaningful ways.

What makes this project especially motivating is that it is inspired by real operational problems experienced by someone close to me. I like the idea of building an application that could genuinely help my sister-in-law better manage customer orders, payments, deliveries, and business records. Knowing that the project could provide practical value to a real person and potentially other small home-run businesses makes the project feel meaningful rather than just being another academic assignment.

I am also excited about the technical challenges involved in the project. The application will allow me to continue building my skills with Angular, Node.js, Express, and MongoDB while also learning completely new concepts such as Progressive Web Application (PWA) development, spreadsheet export functionality, downloadable reporting systems, and Go backend services. I enjoy learning new technologies through hands-on development, and this project gives me the opportunity to expand both my technical skills and my understanding of real-world software architecture.

I understand that senior projects can become difficult and involve technical roadblocks, debugging challenges, and time management difficulties. However, because this project aligns closely with my personal interests, career goals, and desire to create software that helps real people solve problems, I believe I will remain motivated to continue improving and completing the project even when challenges arise.

Milestones, Tasks and Schedule

I have divided the Home Gig project into multiple development phases focused on planning, architecture, backend development, frontend development, Progressive Web Application (PWA) functionality, spreadsheet export functionality, testing, deployment preparation, and operational workflow integration. The project schedule has been designed across both CSE 499A and CSE 499B to provide a realistic development timeline for a solo developer while still allowing flexibility for learning new technologies and refining the project scope as development progresses. For CSE 499A, my schedule is set up to conclude by the end of Week 11 as discussed and approved by my instructor at the beginning of the semester.

Although the project schedule spans two semesters as a fallback plan, I intend to work toward completing a functional minimum viable product earlier if development progress allows. The two-semester schedule is intended to provide sufficient time for research, testing, debugging, refinement, and documentation while maintaining realistic expectations for project completion.

CSE 499A Schedule

Milestone / Task	Estimated Hours	Target Completion
Milestone - Project Proposal	8	Week 4 / May 16, 2026
Requirements gathering and workflow planning	6	Week 5 / May 23, 2026
Milestone - UI/UX wireframes and dashboard layout planning	8	Week 6 / May 30, 2026
Milestone - UI/UX Design Complete	2	Week 6 / May 30, 2026
Milestone - System architecture and database schema design	6	Week 6 / May 30, 2026
Development environment setup and project configuration	4	Week 7 / June 6, 2026
Angular frontend foundation and routing	6	Week 7 / June 6, 2026
Node.js / Express backend API foundation	6	Week 7 / June 6, 2026
MongoDB and Mongoose model setup	5	Week 8 / June 13, 2026
Authentication and authorization prototype	6	Week 8 / June 13, 2026
Product and inventory prototype	6	Week 8 / June 13, 2026
Milestone - Technology Prototype Demo	2	Week 8 / June 13, 2026
Centralized order management workflow	7	Week 9 / June 20, 2026
Manual/verbal order entry workflow	5	Week 9 / June 20, 2026
Payment and delivery status tracking	6	Week 10 / June 27, 2026
Spreadsheet export prototype	5	Week 10 / June 27, 2026
PWA setup and responsive mobile testing	5	Week 11 / July 4, 2026
Requirements Specification documentation	7	Week 11 / July 4, 2026
Milestone - Requirements Specification	2	Week 11 / July 4, 2026

Estimated CSE 499A Total: 102 hours

CSE 499B Schedule

Milestone / Task	Estimated Hours	Target Completion
Review remaining CSE 499A work and adjust scope	2	Week 1
Refine owner dashboard workflow	3	Week 2

Milestone - Improve frontend/backend integration	3	Week 3
Improve spreadsheet export and reporting	3	Week 4
Milestone - Testing and debugging core workflows	4	Week 5
Deployment preparation	3	Week 6
User Acceptance Testing (UAT) preparation	3	Week 7
Milestone - User Acceptance Testing (UAT)	4	Week 8
Apply UAT feedback and usability improvements	4	Week 9
Final bug fixes and workflow polish	4	Week 10
Milestone - Final documentation and portfolio	3	Week 11
Final presentation/demo preparation	3	Week 12
Final demo rehearsal and cleanup	2	Week 13
Milestone - CSE 499B End-of-Semester Demo	1	Week 14

Estimated CSE 499B Total: 42 hours

Accelerated Best-Case Scenario Plan

The following schedule represents an accelerated best-case scenario plan intended to complete a functional minimum viable product (MVP) by the end of Week 11 of CSE 499A. This schedule compresses the projected development effort into a single semester by increasing weekly development hours and prioritizing completion of the core operational workflows required for the Home Gig platform.

At the time of proposal submission, approximately 27 project hours have already been completed, including proposal preparation, project planning, concept refinement, workflow analysis, competitive research, and technology evaluation. To complete the project within the accelerated timeline, the remaining development effort will require approximately 99 additional hours between Weeks 5–11.

Accelerated Best-Case Scenario Schedule

Milestone / Task	Estimated Hours	Target Completion
Milestone - Project Proposal	8	Wk 4 / May 16, 2026
Requirements gathering and workflow planning	6	Wk 5 / May 23, 2026
Milestone - UI/UX wireframes and dashboard layout planning	6	Wk 5 / May 23, 2026
Milestone - UI/UX Design Complete	2	Wk 5 / May 23, 2026
Milestone - System architecture and database schema design	5	Wk 5 / May 23, 2026
Development environment setup and project configuration	4	Wk 6 / May 30, 2026
Angular frontend foundation and routing	6	Wk 6 / May 30, 2026
Node.js / Express backend API foundation	6	Wk 6 / May 30, 2026
MongoDB and Mongoose model setup	5	Wk 6 / May 30, 2026

Authentication and authorization prototype	6	Wk 7 / June 6, 2026
Product and inventory prototype	6	Wk 7 / June 6, 2026
Milestone - Technology Prototype Demo	2	Wk 7 / June 6, 2026
Centralized order management workflow	7	Wk 7 / June 6, 2026
Manual/verbal order entry workflow	5	Wk 8 / June 13, 2026
Payment and delivery status tracking	6	Wk 8 / June 13, 2026
Spreadsheet export prototype	5	Wk 8 / June 13, 2026
PWA setup and responsive mobile testing	5	Wk 8 / June 13, 2026
Frontend/backend integration testing and debugging	6	Wk 9 / June 20, 2026
User interface refinement and workflow improvements	4	Wk 9 / June 20, 2026
Testing and debugging core workflows	4	Wk 9 / June 20, 2026
User Acceptance Testing (UAT) preparation	3	Wk 9 / June 20, 2026
Milestone - User Acceptance Testing (UAT)	4	Wk 9 / June 20, 2026
Apply UAT feedback and usability improvements	3	Wk 10 / June 27, 2026
Requirements Specification documentation	6	Wk 10 / June 27, 2026
Milestone - Requirements Specification	2	Wk 10 / June 27, 2026
Final documentation and portfolio preparation	3	Wk 10 / June 27, 2026
Final presentation/demo preparation	3	Wk 10 / June 27, 2026
Final demo rehearsal and cleanup	2	Wk 11 / July 4, 2026
Final testing, bug fixes, and MVP refinement	6	Wk 11 / July 4, 2026
Milestone - Functional MVP Completion Goal	2	Wk 11 / July 4, 2026
Milestone - Final Documentation and Portfolio	1	Wk 11 / July 4, 2026
Milestone - CSE 499B End-of-Semester Demo	1	Wk 11 / July 4, 2026

Accelerated Plan Estimated Total

Category	Hours
Previously Completed Hours	19
Remaining Scheduled Hours	135
Total Project Hours	154

Resources

The Home Gig project will primarily use existing personal computer hardware along with free and open-source software technologies. The project will also rely on online documentation, tutorials, reference materials, and mentorship resources to support learning and development throughout the semester.

Hardware Resources

Resource	Purpose	Estimated Cost
Personal Desktop Computer	Primary development environment for frontend, backend, database, and testing workflows	Already Owned

Mobile Devices	Testing responsive design and Progressive Web Application (PWA) functionality	Already Owned

Software and Development Tools

Resource	Purpose	Estimated Cost
Visual Studio Code	Primary code editor and development environment	Free
Angular Framework	Frontend web application framework	Free
Node.js	Backend runtime environment	Free
Express.js	Backend API framework	Free
MongoDB Community Edition	Database management system	Free
Mongoose	MongoDB object modeling library	Free
Bootstrap	Responsive user interface framework	Free
Git	Version control system	Free
GitHub	Source code hosting and repository management	Free
ExcelJS or SheetJS	Spreadsheet export and reporting functionality	Free
Go (Golang)	Backend reporting and spreadsheet service experimentation	Free
GitHub Copilot	AI-assisted code completion, code suggestions, and software development assistance within Visual Studio Code	Free Version
ChatGPT	AI-assisted research, debugging, architectural planning, troubleshooting, learning support, and development assistance	$20/month Subscription
Codex	AI-assisted software engineering and coding support integrated with the ChatGPT subscription	Included with ChatGPT Subscription
SuperGrok	AI-assisted research, troubleshooting, brainstorming, and software development support	$20/month Subscription
Agent Skills Plugins and Intelligent Development Assistance Tools	AI-assisted task automation, workflow orchestration, research support , debugging support, and software development experimentation	Free / Existing Subscription

Learning Resources and Reference Materials

Resource	Purpose	Estimated Cost
Official Angular Documentation	Learning PWA development and Angular features	Free

Official Node.js and Express Documentation	Backend API and server development reference	Free
MongoDB Documentation	Database design and query reference	Free
Go Documentation and Tutorials	Learning foundational Go programming concepts	Free
YouTube Tutorials and Online Videos	Learning new technologies and troubleshooting	Free
Stack Overflow and Technical Forums	Research and debugging assistance	Free

Mentors and Professional Resources

Resource	Purpose	Estimated Cost
Rob Tuft	Project mentorship, technical guidance, and feedback	Volunteer / No Cost
Ken Lance	Project mentorship, technical guidance, and feedback	Volunteer / No Cost
Chris Adzima	Project mentorship, technical guidance, and feedback	Volunteer / No Cost

<u>Estimated Total Cost</u>
Most technologies and tools selected for this project are free and open-source. Since existing hardware will be used and no paid software licenses are currently required, the estimated direct project cost is minimal.

<u>Estimated Total Project Cost: $0 – $100</u>
This estimate allows for possible incidental costs such as optional hosting services, testing tools, or small development resources if needed later in the project.

Dependencies

The Home Gig project depends on several software technologies, development tools, hardware resources, and learning resources in order to successfully design, develop, test, and deploy the application. The project will primarily use a full-stack web development architecture based on Angular, Node.js, Express, and MongoDB, along with additional technologies required for Progressive Web Application (PWA) support and spreadsheet export functionality.

<u>Programming Languages and Frameworks</u>
The project will require the installation and use of the following programming languages, frameworks, libraries, and technologies:

- TypeScript
- JavaScript
- HTML
- CSS
- Go (Golang)

- Angular
- Node.js
- Express.js
- MongoDB
- Mongoose
- Bootstrap
- ExcelJS or SheetJS
- Git and GitHub

<u>Development Environment and IDE</u>
The primary development environment and IDE for the project will be:
- <u>Visual Studio Code</u>

Additional development tools may include:
- Angular CLI
- Node Package Manager (npm)
- MongoDB Compass
- Git Bash or terminal tools
- Chrome Developer Tools

<u>Platforms and Operating Systems</u>
The application will primarily be developed and tested on:
- Windows 10 desktop environment

The final application itself will be:
- A web-based application
- Mobile responsive
- Installable as a Progressive Web Application (PWA)

The application will be accessible through:
- Desktop web browsers
- Mobile web browsers
- Installed PWA environments on mobile devices

<u>Development and Testing Environment</u>
Development and testing will primarily occur:
- On a personal desktop computer
- Using local development servers
- Using MongoDB local database instances
- Through browser-based testing
- Through mobile device testing for responsive and PWA functionality

Testing will include:
- Frontend user interface testing
- Backend API testing

- Database integration testing
- Spreadsheet export testing
- Mobile responsiveness testing
- PWA installation and behavior testing

Installation and Deployment

The project will initially be developed and tested locally using:

- Angular development server
- Node.js and Express backend server
- MongoDB Community Edition

Deployment may involve:

- GitHub repository hosting
- Web hosting platforms such as Vercel, Netlify, Render, or similar services
- Possible cloud database hosting if required

The exact final deployment method may change depending on project requirements and available free hosting options.

External Dependencies and Assistance

The project is largely self-contained and does not depend on another developer providing required software components. However, the project will rely on:

- Open-source libraries and frameworks
- Online technical documentation
- Community tutorials and reference materials
- Mentorship and technical guidance from Rob Tuft, Ken Lance and Chris Adzima

These mentors may provide:

- Feedback on project architecture
- Advice on software engineering practices
- Technical troubleshooting assistance
- General project guidance

Permissions and Approvals

At this time, no special permissions, licenses, or legal approvals are expected to be required for the project. The project will use primarily free and open-source technologies and development tools. If external hosting services or third-party integrations are later introduced, additional account setup or configuration permissions may be required.

Risks

Several risks and challenges may affect the successful completion of the Home Gig project. The project involves multiple technologies and software engineering concepts that I have limited or no prior experience with, which creates both technical and scheduling risks. However, identifying these risks early will help me better prepare for challenges and manage the project scope appropriately throughout development.

One of the largest risks is learning Progressive Web Application (PWA) development. While I already have beginner-level experience with Angular, Node.js, Express, and MongoDB, I currently have no experience building installable PWAs. I will need to learn concepts such as service workers, caching strategies, application manifests, offline support, and mobile installation behavior. Troubleshooting cross-browser and mobile compatibility issues may also require additional research and testing time.

Another significant risk involves spreadsheet export and downloadable reporting functionality. I have no prior experience generating downloadable CSV or Excel files within a web application. To implement this feature successfully, I will need to learn how to use libraries such as ExcelJS or SheetJS, generate properly formatted spreadsheet data, handle file streaming, and manage browser-based downloads securely and reliably.

The project also introduces the additional challenge of learning foundational Go programming concepts. Since I do not currently have experience with the Go programming language, implementing a small backend reporting or spreadsheet generation service may require more time than anticipated. There is also a risk that integrating a secondary backend language into the project architecture could increase complexity and debugging difficulty.

Another project risk involves managing the overall project scope within the available timeline. The project contains several interconnected features including:

- User authentication
- Product and inventory management
- Centralized order workflows
- Manual/verbal order entry
- Payment tracking
- Delivery tracking
- Spreadsheet exports
- Responsive mobile design
- Progressive Web Application (PWA) functionality

Because all of these systems interact with each other, delays in one area may impact progress in others. To reduce this risk, the project is intentionally scoped as a minimum viable product rather than a large enterprise-scale platform.
Additional risks include:

- Debugging frontend and backend integration issues
- Managing asynchronous API communication
- Learning deployment and hosting workflows
- Mobile device compatibility testing
- Balancing project work with other coursework and responsibilities
- Underestimating the time required to learn unfamiliar technologies

There is also a possibility that some advanced or optional features may need to be simplified, postponed, or reduced if development takes longer than expected. If necessary, priority will be placed on completing the core operational workflows of the application, including centralized order management, manual order entry, payment tracking, and spreadsheet export functionality.

Despite these risks, I believe this project is achievable because many of the core technologies are already somewhat familiar to me, and the project scope has been intentionally designed to remain flexible and scalable based on development progress. The project also provides a valuable opportunity to strengthen self-guided learning, troubleshooting, research, and software engineering problem-solving skills.